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Development of a Network-based Timing and Tag Information Distribution System for Synchrotron Radiation Experiments at SPring-8

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Outline

- Purpose of this study
- Building the Proof of Concept (PoC) System
 - Evaluation of the system
- Improvement of the PoC System
 - Evaluation of the improved system
- Future Plan
- Summary

Purpose of this study

- Time-resolved measurements in synchrotron radiation experiments requires high-precision timing signals:
 - RF Clock; synchronized with a storage-ring accelerator ; **~508.58 MHz@SPring-8**
 - Fundamental Revolution Frequency (Zero Address) **Signal**; **~208.8 kHz@SPring-8**

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 - Fundamental Revolution Frequency (Zero Address) **Signal**; **~208.8 kHz@SPring-8**
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 - long distance RF cable,
 - various types of electronics; Ex.) Frequency divider module, Timing delay module.

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- These timing signals have been introduced into the beamline by deploying
 - long distance RF cable,
 - various types of electronics; Ex.) Frequency divider module, Timing delay module.
- Difficulties in the ***“analog signal distribution”***;
 - These installations are costly, and require significant effort to adjust the timing signal.
 - Beamlines that can utilize these precise timing signals are restricted
 - because their expansion is difficult.
 - Difficult to include additional information such as a shot number, timestamp.

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→ Study a timing distribution system based on a high-precision time synchronization technology over a standard network.

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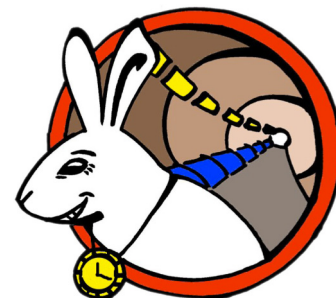
→ Study a timing distribution system based on a high-precision time synchronization technology over a standard network.

- The system delivers “**digital information**” related to timing;
 - Timing signal can be handled and expanded without difficulty,
 - Timing signal can be precisely and easily adjusted,
 - Other useful information can be attached easily.
 - Time stamp, Shot number, ...

Purpose of this study

→ Study a timing distribution system based on **a high-precision time synchronization technology** over a standard network.

- adopt **White Rabbit (WR)**
 - is promoted as an international collaborative project.
 - **extends IEEE 1588 standard.**
 - **achieves sub-nanosecond accuracy** by using
 - Synchronous Ethernet,
 - precise phase measurement by DDMTD (Digital Dual Mixer Time Difference).
 - **ensures deterministic data transfer.**
- opened to the public at the Open Hardware Repository (OHR);
 - <https://www.ohwr.org/projects/white-rabbit>

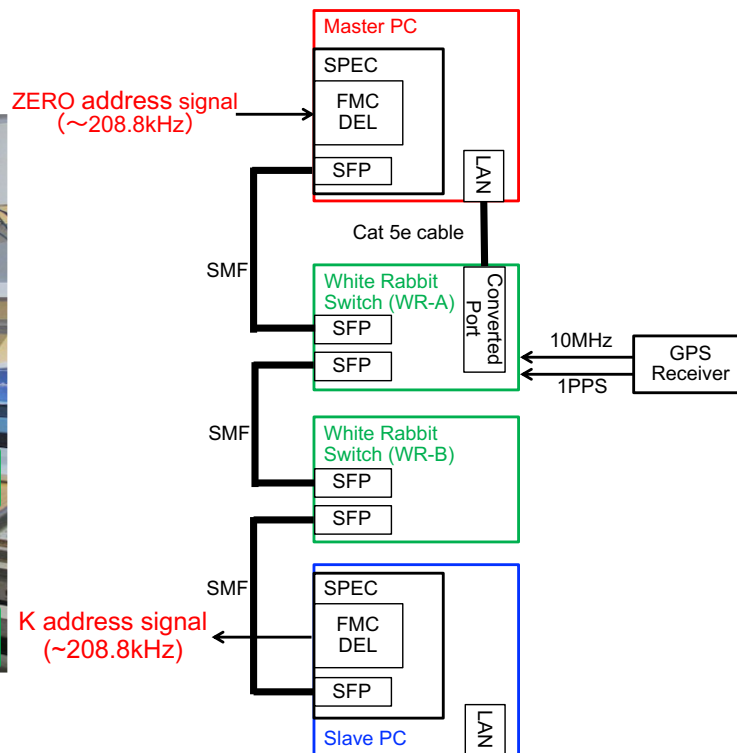
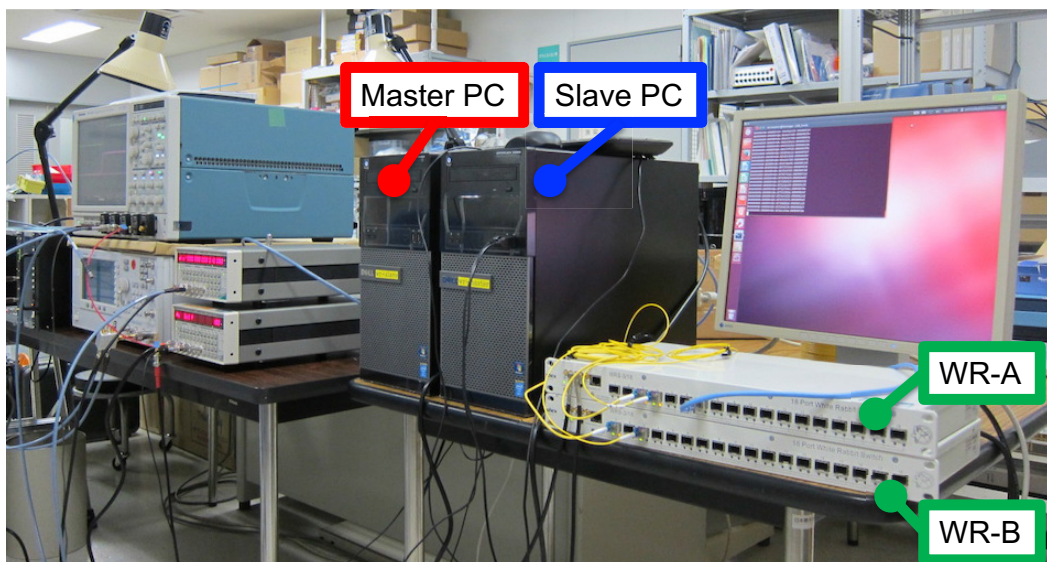


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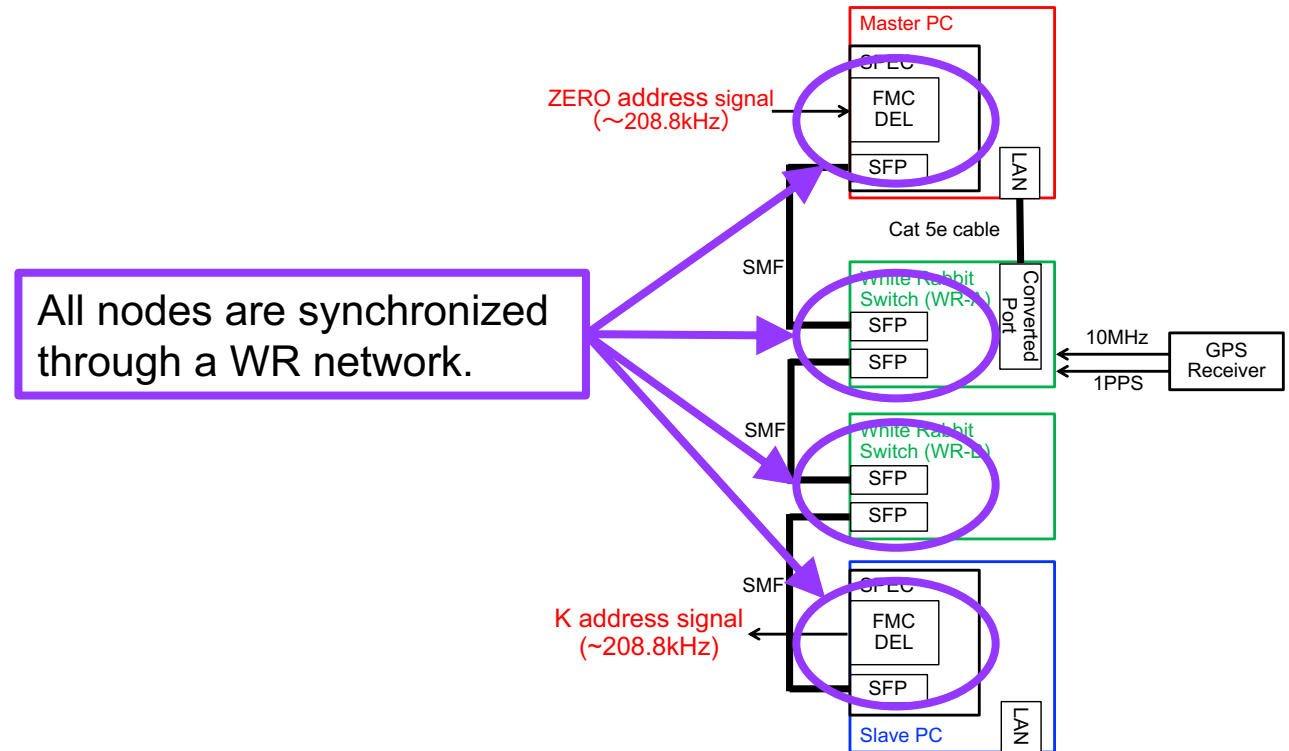
→ Study a timing distribution system based on a high-precision time synchronization technology over a standard network.

- Build the proof of concept (PoC) system to realize;
 - generation of 208.8 kHz revolution frequency signal synchronized with the target bunch of the SPring-8 storage ring (accelerated by 508.58 MHz RF),
 - delivery of tag information.

Building the PoC system

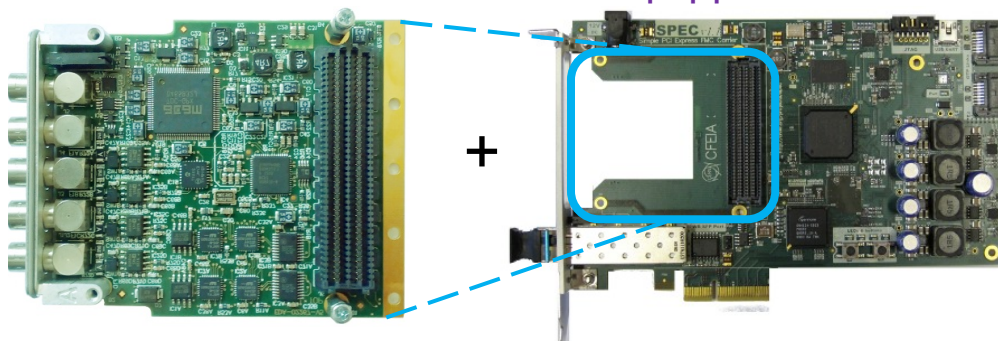


Building the PoC system



Building the PoC system

Master & Slave PCs are equipped with



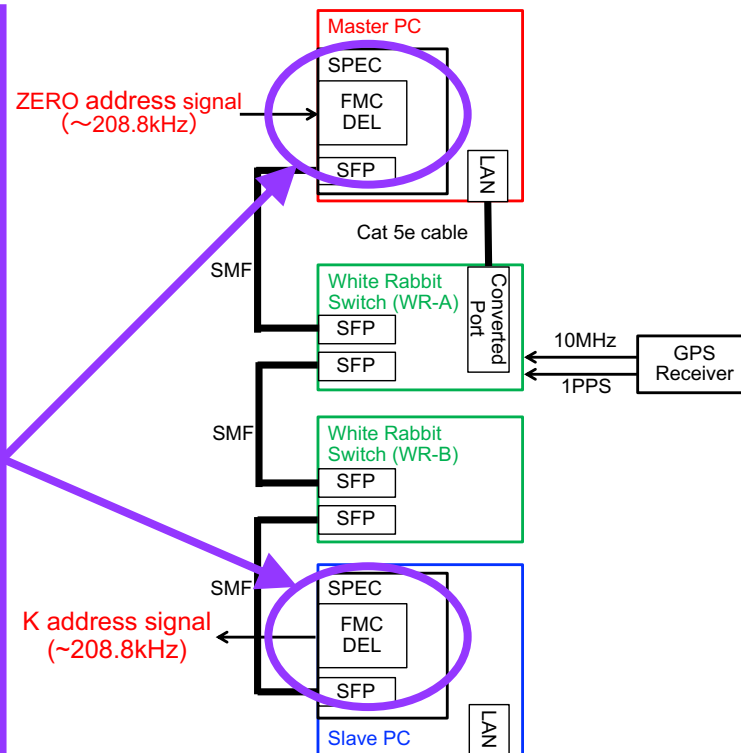
FMC DEL 1ns 4cha card
(FMC DEL)

- trigger input channel : 1
- output channels with PD : 4
- Resolution : 10ps
- Operation mode;
 - pulse delay mode
 - pulse generator mode
 - TDC mode

PCIe FMC carrier board
(SPEC)

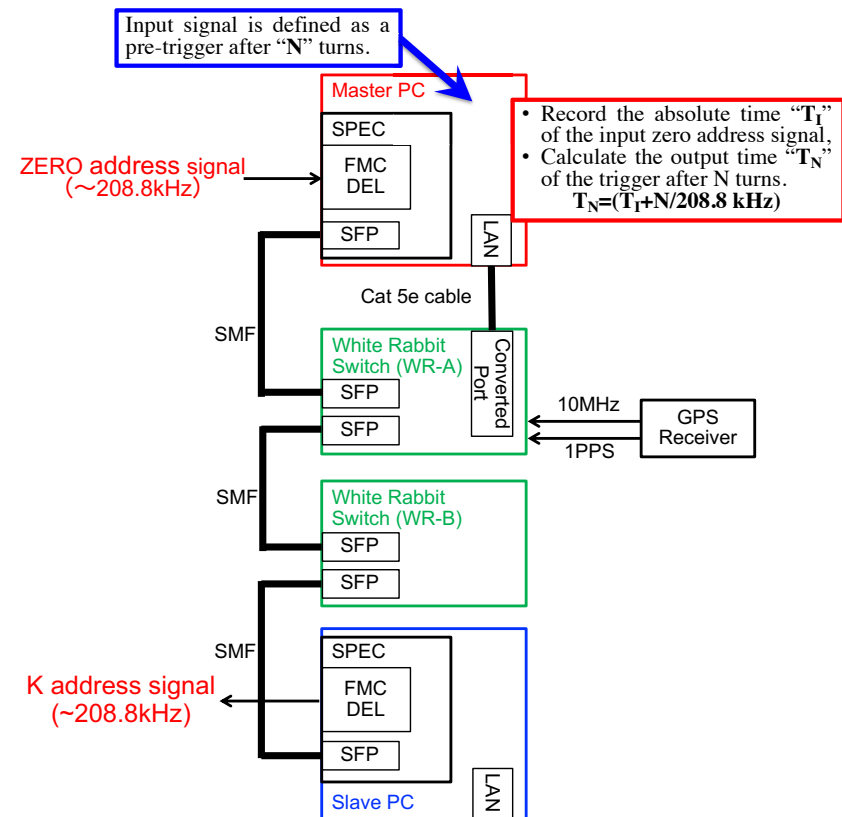
- Xilinx Spartan-6 FPGA
- PCIe x4
- FMC slot with LPC connector
- SFP cage x1

also opened to the public at OHR



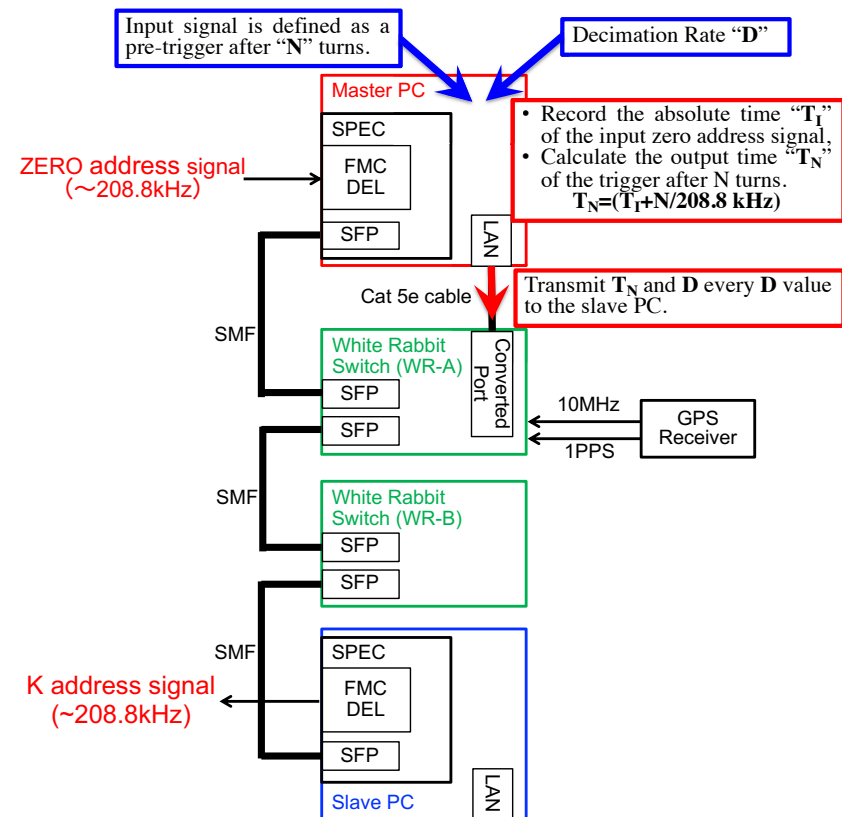
Building the PoC system

- programmable parameters;
 - N (pre-trigger after N turns)



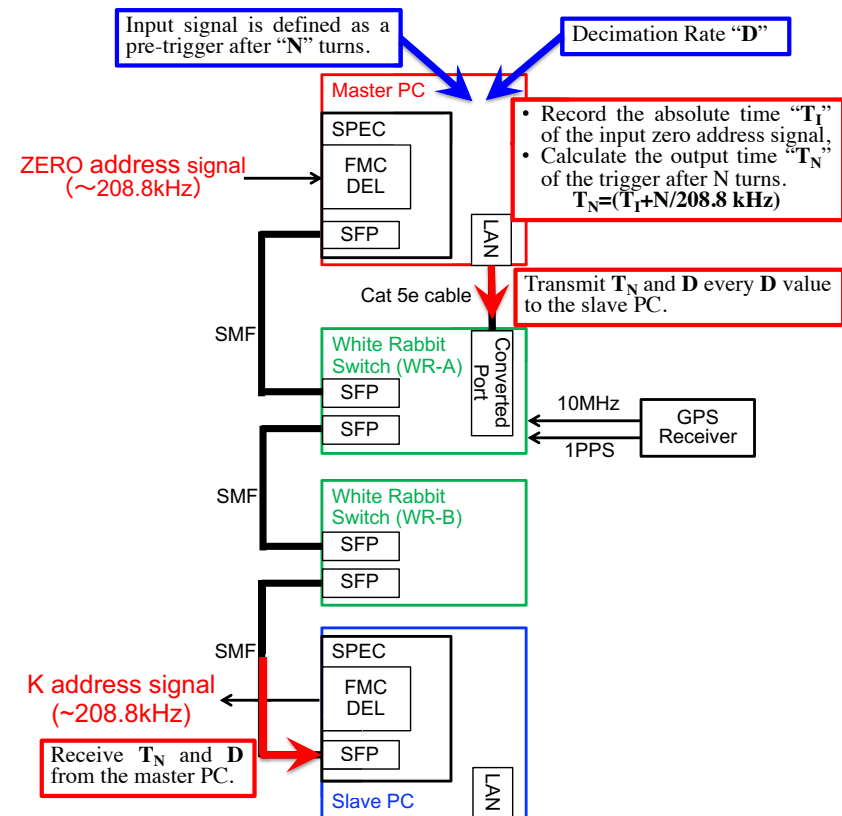
Building the PoC system

- programmable parameters;
 - N (pre-trigger after N turns)
 - D (decimation rate)



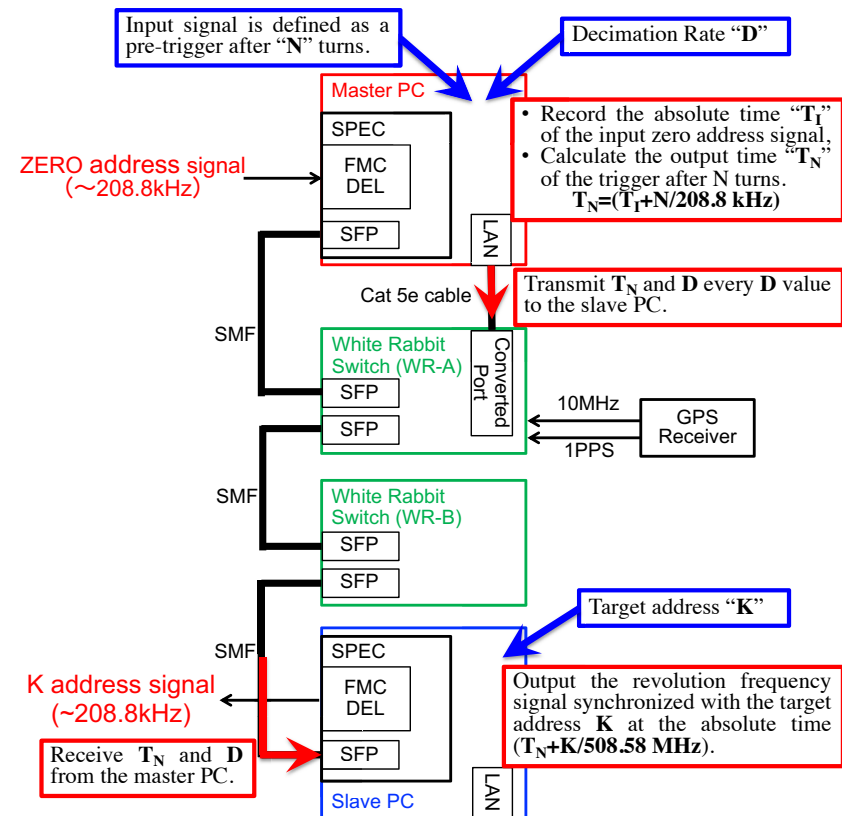
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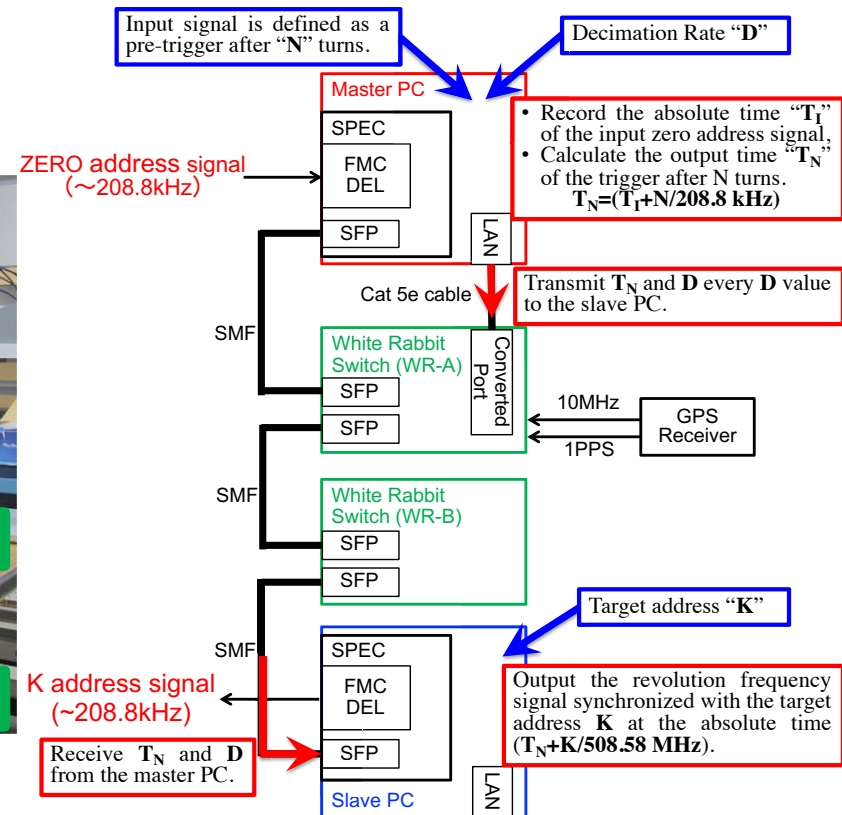
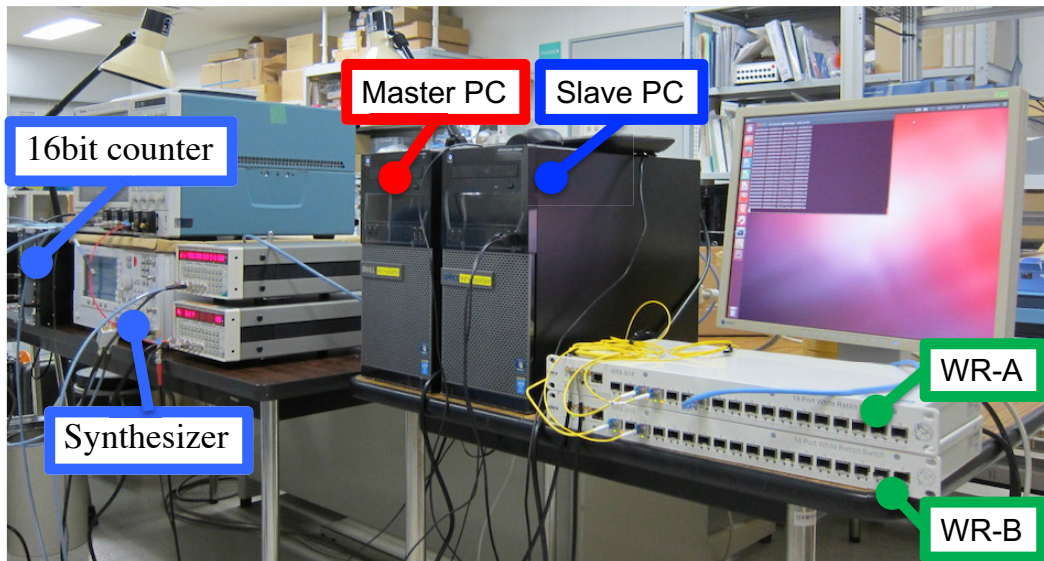


Building the PoC system

- programmable parameters;
 - N (pre-trigger after N turns)
 - D (decimation rate)
 - K (Target bunch address)



Evaluation of the PoC system

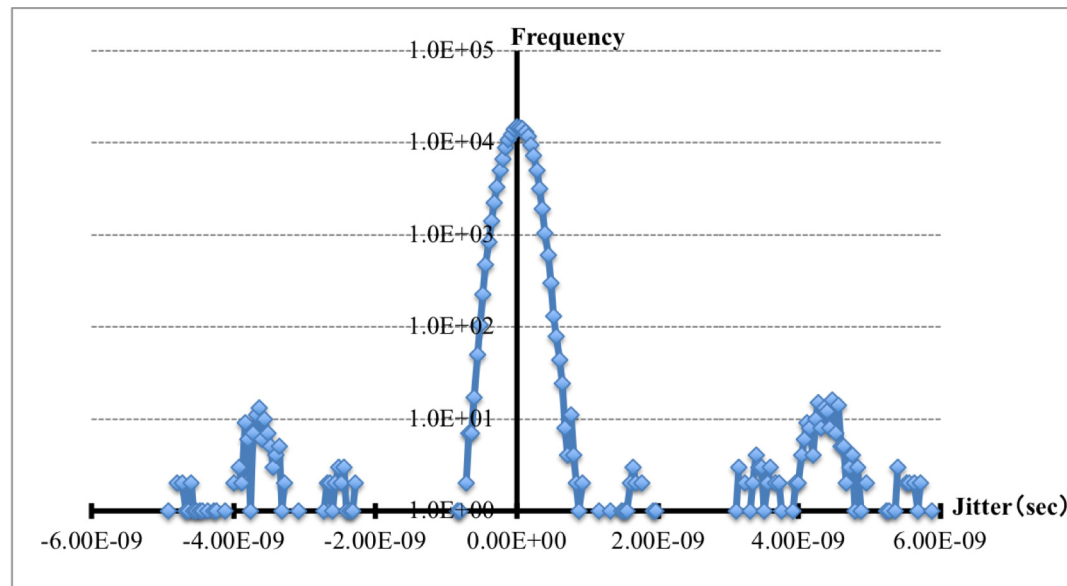


Evaluation of the PoC system

- Jitter measurement of the 208.8 kHz output signal.
 - $D = 1000, N = 900$

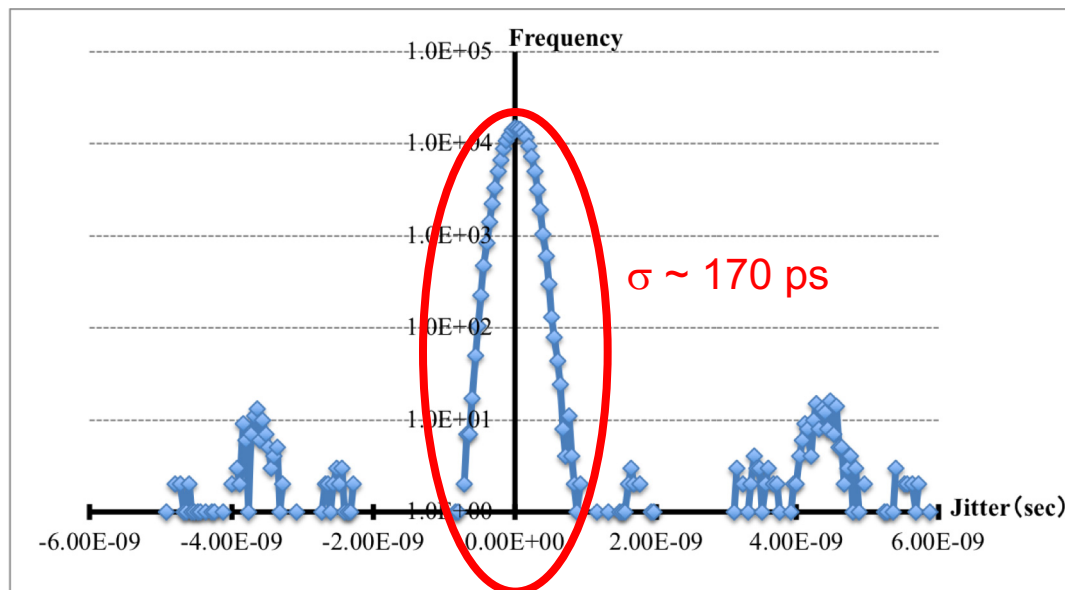
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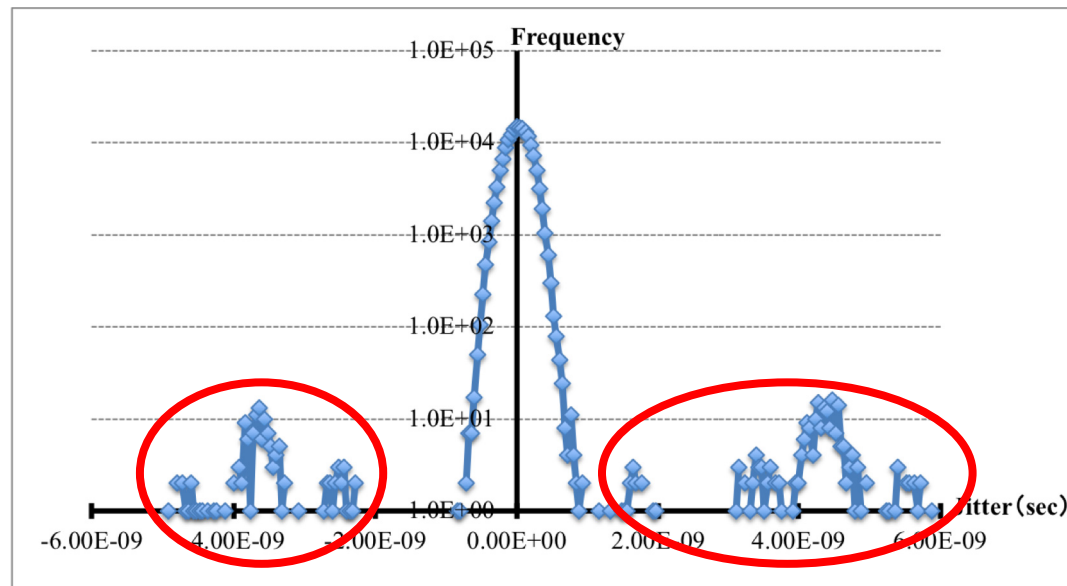
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 - $D = 1000, N = 900$
- jitter of the center structure was ~ 170 ps (1σ).
- Observed unnatural structures,
 - large jitter beyond ± 2 ns.



Evaluation of the PoC system

- Measurement of the decimation rate D dependence of jitter.

Evaluation of the PoC system

- Measurement of the decimation rate D dependence of jitter.
 - Jitter decreases with decrease in D as expected.
 - To lower the jitter, D should be reduced as much as possible.

Parameters	Jitter of center structure (1σ)	Entire Jitter (1σ)	Number of samples
$D = 1000, N = 900$	171 (ps)	254 (ps)	163 k
$D = 600, N = 500$	139 (ps)	221 (ps)	68 k
$D = 300, N = 200$	131 (ps)	201 (ps)	71 k

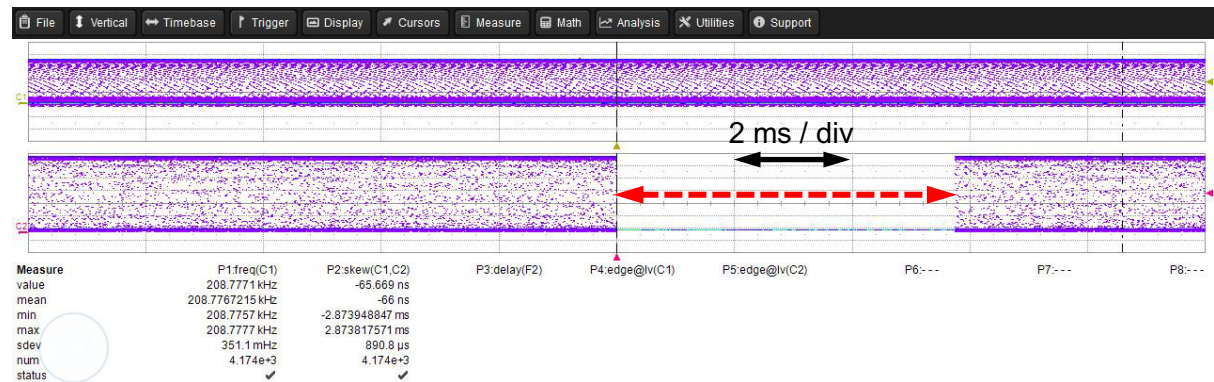
Evaluation of the PoC system

- Measurement of the decimation rate D dependence of jitter.
 - Jitter decreases with decrease in D as expected.
 → To lower the jitter, D should be reduced as much as possible.
- Only the case of $D > N$ was implemented to simplify the FPGA logic construction...

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Evaluation of the PoC system

- Problems of the PoC system
 1. Large jitter beyond ± 2 ns,
 2. There was the limitation of $D > N$ in the operation parameters,
 3. Output signal from the slave was sometimes missing.

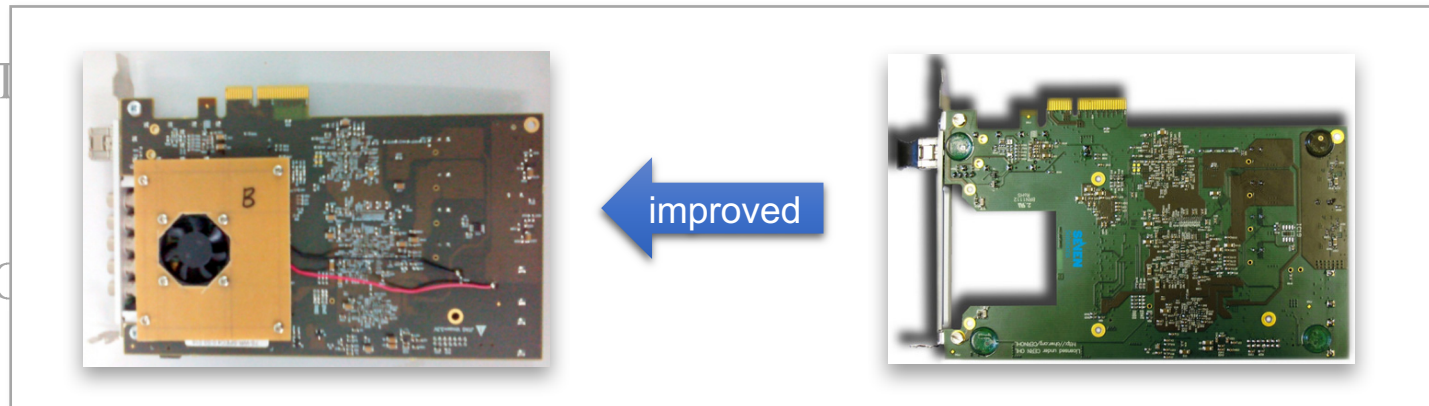


Improvement of the PoC system

- The PoC system has been improved to solve the problems.
 1. Large jitter beyond ± 2 ns
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 3. Output signal from the slave was sometimes missing.

Improvement of the PoC system

- The PoC system has been improved to solve the problems.
 1. Large jitter beyond ± 2 ns
 - Caused by heating up of the FMC DEL.
 - Core temperature of the FMC DEL surpassed over 70 °C.
 - The heat problem of the FMC DEL was documented on the OHR website.
→ Attach a small cooling fan on the SPEC board to cool the FMC DEL continuously according to the design guide provided on the OHR.



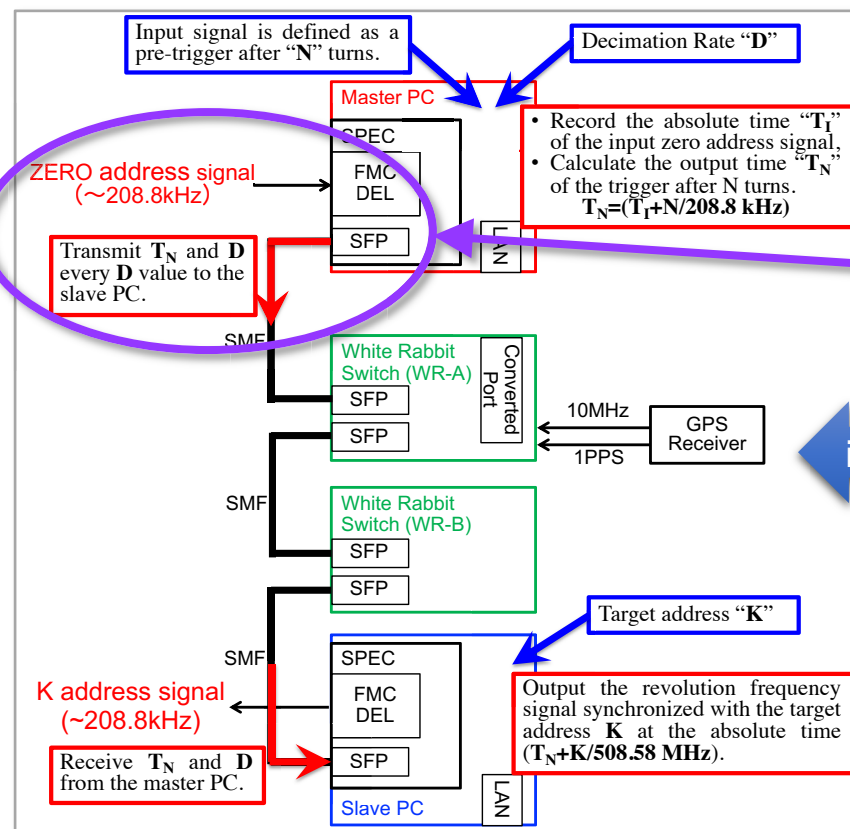
Improvement of the PoC system

- The PoC system has been improved to solve the problems.
 1. Large jitter beyond ± 2 ns
 - Attach a small cooling fan on the SPEC board.
 2. There was the limitation of $D > N$ in the operation parameters
 - the logic for the case of $D < N$ has been newly developed, and successfully integrated into the system.
 3. Output signal from the slave was sometimes missing.

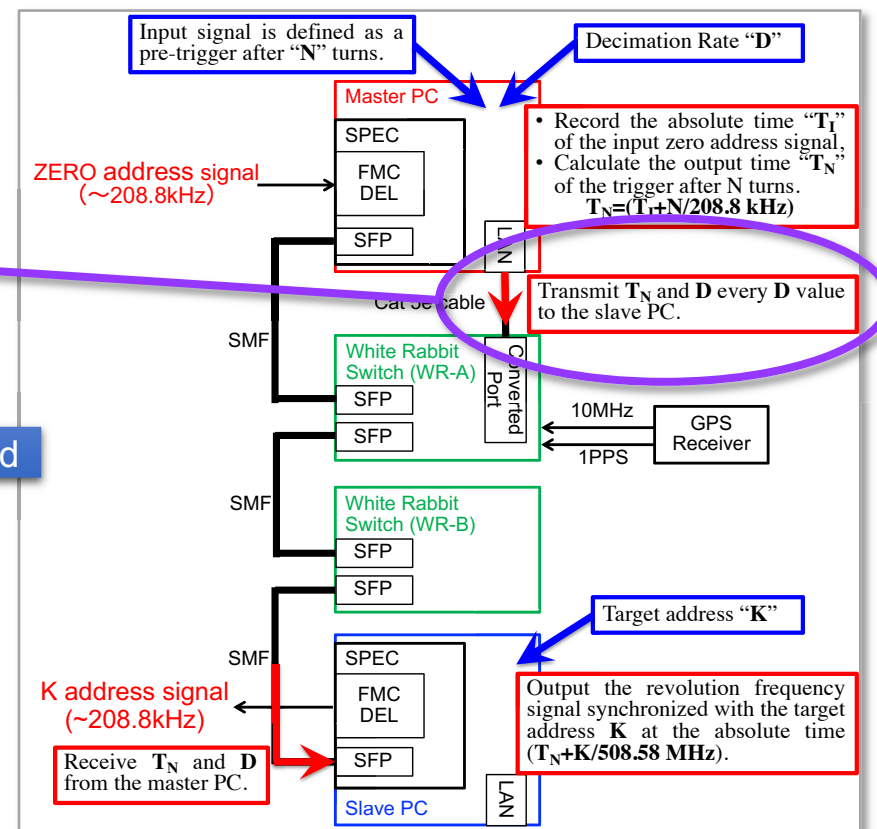
Improvement of the PoC system

- The PoC system has been improved to solve the problems.
 1. Large jitter beyond ± 2 ns
 → Attach a small cooling fan on the SPEC board.
 2. There was the limitation of $D > N$ in the operation parameters
 → Integrate the new logic for the case of $D < N$.
 3. Output signal from the slave was sometimes missing.
 - Caused by an execution delay of the software
 - Responsible for transmitting the output time T_N and decimation rate D .
 - Employ “**Etherbone master core**” to realize data transmission by the FPGA on the master side as originally planed.
 - “**Etherbone master core**” was a missing piece at the beginning of this project,
 - Fortunately, “**Etherbone master core**” was released at this improvement timing.

Improvement of the PoC system



improved



Improvement of the PoC system

- The PoC system has been improved to solve the problems.
 1. Large jitter beyond ± 2 ns
→ Attach a small cooling fan on the SPEC board.
 2. There was the limitation of $D > N$ in the operation parameters
→ Integrate the new logic for the case of $D < N$.
 3. Output signal from the slave was sometimes missing.
→ Realize automatic data transmission by the FPGA.

Evaluation of the improved PoC system

- Jitter measurement of the 208.8 kHz output signal.

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Parameters	Jitter (1σ)	Number of samples
D = 1000, N = 1000	155.2 (ps)	108.2 k
D = 50, N = 1000	97.33 (ps)	458.3 k

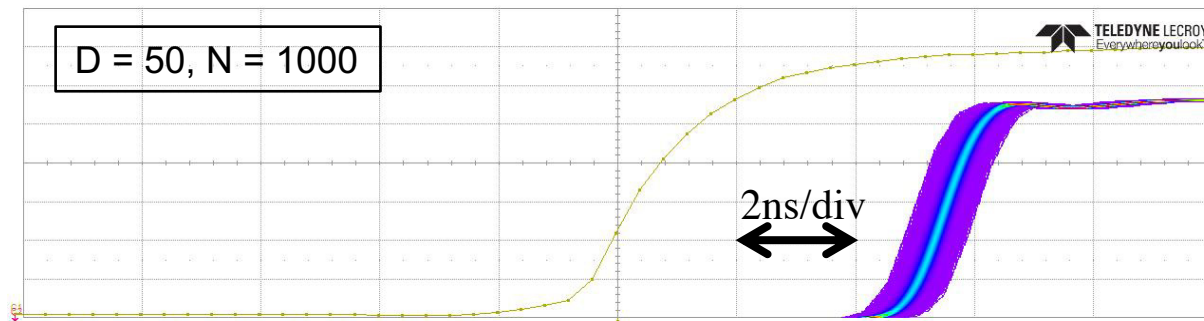
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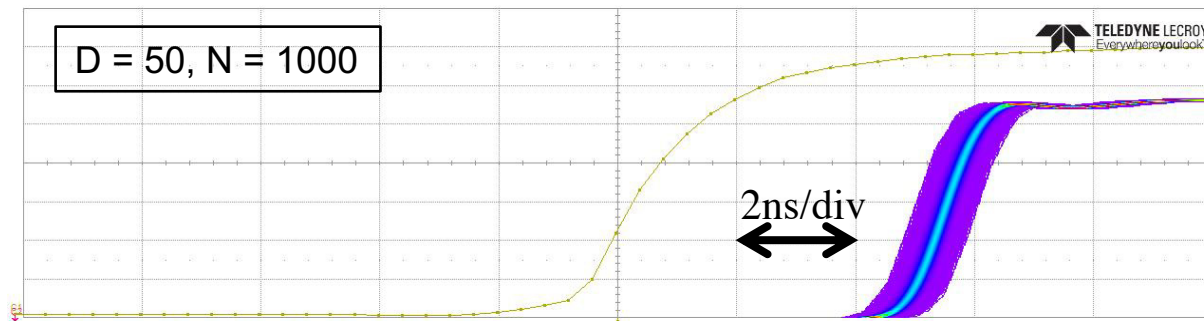
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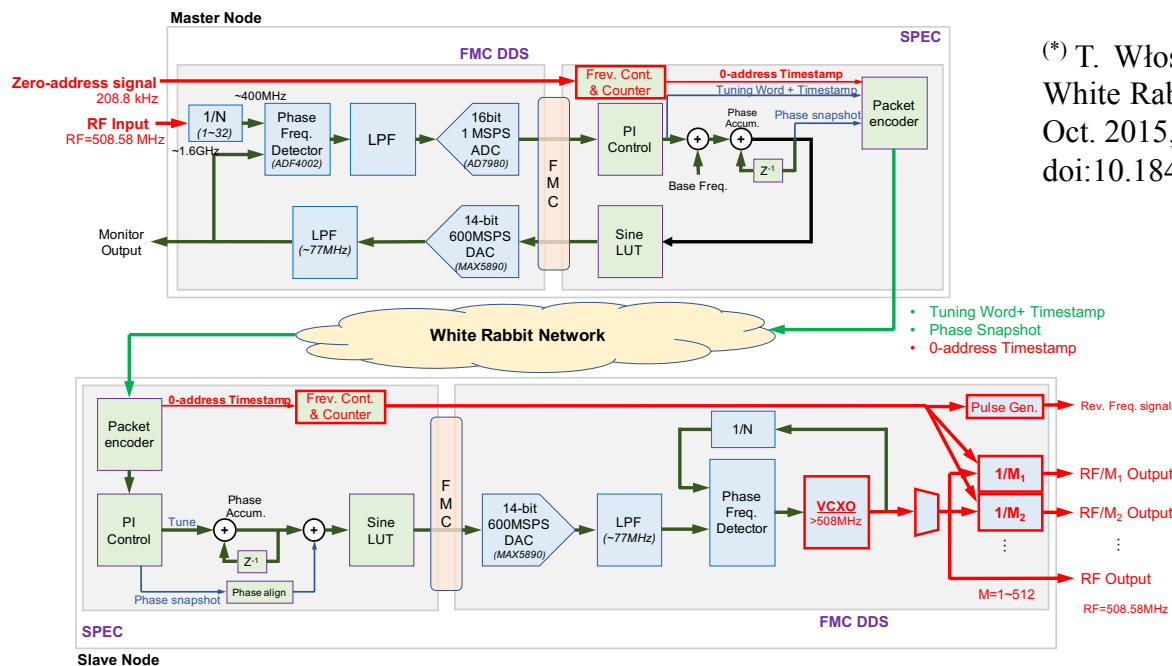
- The large jitters disappeared by cooling the FMC DEL.
- Dropping of the output signal were not observed while the system was working.

Evaluation of the improved PoC system

- Stability of the improved system.
 - There was an occasion where the system ran continuously for over 18 hours, while on another occasion the system stopped immediately.
 - There was an occasion where the slave PC hung up during the test.
 - The system has not become functional after a series of measurements.
 - Replaced all of the hardware with new ones, but the problem has not been fixed yet.
 - have to investigate the cause of the problem and fix it.

Future Plan

- Have commenced the study of the accelerator RF clock (508.58 MHz) distribution over the WR network.
 - Extend a proof of concept system by CERN (*) to the SPring-8 case,
 - Also integrate 208.8 kHz distribution function based on my PoC system into the future system.



(*) T. Włostowski *et al.*, “Trigger and RF Distribution Using White Rabbit”, in Proc. ICALEPCS’15, Melbourne, Australia, Oct. 2015, pp. 619-623.
doi:10.18429/JACoW-ICALEPCS2015-WEC3O01

Summary

- I have developed the PoC system with the aim of building a trigger and tag information distribution system based on a high precision time synchronization technology.
- The initial PoC system has achieved an output jitter of 127 ps (1σ) at the decimation rate $D = 300$.
- The improved PoC system has successfully realized automatic transmission of the information by means of hardware, and accomplished a jitter below 100 ps (97 ps (1σ)).
- However, it has a unresolved severe stability problem.
- I have commenced the study of a 508.58 MHz clock distribution over the WR network. I also plan to integrate the distribution of 208.8 kHz revolution frequency based on the PoC system after fix the problem.